

ABSTRACT:

Name of Responsible Researcher:	Kornelius Kupczik
Proposal Title:	Advancing our understanding of Mesotheriid feeding ecology

Describe the main issues to be addressed: objectives, methodology and expected results. **The maximum length for this section is 1 page** (use letter size format, Verdana font size 10 or similar).

The proposed project builds upon the biostratigraphical description and paleobiological analysis of late Miocene fossil vertebrates from the Cuenca Copaquilla-Ticnamar outcrops in Northern Chile funded by a previous Fondecyt Regular project. Inspired by these findings and subsequent observations with regard to the feeding ecology and morphology of the associated fauna, we aim to provide an in-depth understanding of the paleoecological framework of one of the key fossils from this region, *Caraguatypotherium munozi* Flynn et al. 2005 (Mammalia, Notungulata, Mesotheriidae).

We specifically aim to 1) investigate the masticatory morphology and kinematics of *C. munozi*; 2) reconstruct the paleodiet of *C. munozi*; 3) reveal the paleoenvironmental conditions, *C. munozi* and related fauna were exposed to; and finally, 4) expand the existing hypodigm of *C. munozi* and other Miocene fauna from the Cuenca Copaquilla-Ticnamar outcrops.

We will employ an innovative methodological approach by combining different lines of evidence from craniodental morphology, chewing kinematics, dental wear as well as stable isotope ecology. Moreover, we intend to increase the known fossil sample of *C. munozi* and associated fauna by conducting further fieldwork.

In line with the above aims, 1) virtual three-dimensional (3D) models will be obtained from the isolated teeth and complete tooth rows of *C. munozi* specimens. By using the Occlusal Fingerprint Analyzer (OFA) we will characterize and quantify the wear facet patterns on the molars to reconstruct occlusal 3D jaw movements. Based on previous morpho-functional observations it is expected that *C. munozi* engaged in circular and fine rotational movements to be confirmed by OFA.

2) Next, 3D surface texture analysis will be used to quantify various micrometer sized geometric features on the tooth wear facet surfaces obtained with confocal microscopy. We hypothesize that *C. munozi* preferred soft foods but that tooth wear was driven by the presence of exogenous abrasive particles (e.g. grit or volcanic ash) in the environment and/or on the foods.

3) We will analyze the $\delta^{13}\text{C}$ and $\delta^{18}\text{O}$ composition on the carbonate fraction of tooth enamel samples to assess the feeding behavior and environmental factors related to temperature, seasonality patterns and habitat type. It is expected that specimens of *C. munozi* alongside other local herbivores will show $\delta^{13}\text{C}$ and $\delta^{18}\text{O}$ patterns reflective of a generalized diet based on a variety of forest C3 plants.

4) We will systematically search the Cuenca Copaquilla-Ticnamar outcrops again for fossil material, aiming for further craniodental remains of Mesotheriidae as it has recently been suggested that there may more than one species present at the locality. Moreover, we will collect rock samples for geochronological dating.

The proposed project will provide much needed information on the feeding ecology and palaeoenvironment of Mesotheriidae (Notungulata), the family to which *C. munozi* belongs, from the late Miocene of Chile. The expected findings will furthermore inform on the evolutionary, paleoecological and paleogeographical complexity present in the Central Andes, in a time of high faunal provinciality developing during the last large pulse of the lifting Altiplano. Moreover, the project will provide the opportunity to train future generations of paleontologists, anthropologists and biologists in Chile in state-of-the-art 3D tooth wear and palaeoecological analyses.